Multimodal Machine Learning for Red Supergiant Supernova Progenitor Prediction

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Abstract. This paper presents a multimodal machine learning framework that combines pre-explosion optical imaging with catalog-based stellar parameters to estimate the likelihood of core-collapse events in red supergiant (RSG) stars, which are widely accepted as the dominant progenitors of Type II supernovae.
Astrophysical attributes such as effective temperature, bolometric luminosity, stellar radius, extinction, and mass-loss indicators are obtained from large-scale stellar catalogs, while archival pre-explosion images of confirmed progenitors are extracted from wide-field sky surveys.
A pretrained ResNet-18 convolutional encoder is used to transform image cutouts into fixed-length feature embeddings, which are subsequently fused with tabular stellar descriptors through a feature-level integration pipeline. The resulting multimodal representation is used to train a calibrated classifier that outputs a continuous supernova probability score.
Experimental evaluation on a curated dataset of RSGs and confirmed progenitors shows that multimodal fusion consistently outperforms models based solely on tabular data, yielding higher accuracy, improved F1-scores, and better-calibrated probability estimates. The model demonstrates strong discriminative capability while maintaining a conservative, high-precision operating regime suitable for resource-constrained observational follow-up. These results underline the value of jointly leveraging imaging morphology and stellar parameters for the early identification and prioritization of high-risk supernova progenitor candidates.

Keywords: Supernova prediction · Deep learning · CNN embeddings · Red supergiants · Multimodal fusion · Gaia DR3 · Pan-STARRS1 · Explainable AI

1 Introduction

Red supergiants (RSGs), the final stage of evolution of stars with masses in the range of 8-25 $M_{\odot}$, have been identified through direct identification in pre-explosion imaging and population analyses as the main progenitors of Type IIP supernovae [6].

Even in light of these advancements, few RSGs have been definitively seen to have exploded, and the properties that mark stable RSGs from those that are about to experience core collapse are poorly known [6,7]. This is made further complicated by complicated final stages like increased mass loss, convective envelopes, circumstellar dust formation, and eruptive events [1,3,7].

Supernovae are temporary but very luminous star explosions, and their importance in the process of chemical enrichment and the creation of compact remnants, as well as neutron stars and pulsars, cannot be overstated. As is pointed out in the NASA review *The Supernova: A Stellar Spectacle* by author L. Straka [12], "To understand supernovae, one has to integrate spectral observations, models, and multi-wavelength data." Yet, as stated, some features of these processes are not fully understood and hence, data-driven models relating antecedent properties of the star, possibly evidenced in precursory transients, are of considerable interest.

Synoptic surveys like Pan-STARRS1 and Zwicky Transient Facility routinely image massive stars before the event, while Gaia Data Release 3 offers high-precision astrometric, multi-photometry, and calculated astrophysical properties of millions of stellar objects in the Galaxy [4,5,8,9]. These complementing sources naturally call for multimodal machine learning tasks, which attempt joint analysis of imaging data together with star catalog data in estimating supernova likelihoods.

Most existing machine learning studies in transient astronomy focus on post-explosion classification of light curves or difference-image cutouts. Frameworks such as Superphot++ and SuperNNova demonstrate strong performance in this regime but do not address progenitor-level prediction or pre-explosion forecasting [13,14]. Consequently, there is currently no widely adopted method for estimating explosion likelihoods directly from static pre-explosion observations.

This work addresses this gap by developing a multimodal framework that integrates pre-explosion imaging with stellar catalog parameters to estimate explosion likelihoods for red supergiants. Although the approach does not yet provide explicit time-to-event predictions, it shifts the predictive focus upstream toward progenitor properties and establishes a reproducible foundation for future temporally resolved models.

The remainder of this paper is organized as follows. Section 2 reviews related work, Section 3 describes the proposed methodology, Section 4 details the datasets and feature representations, Section 5 presents the results and discussion, and Section 6 concludes with directions for future research.

2 Literature Survey

Deep learning techniques have proved to be effective for modeling complex high-dimensional data in a variety of scientific fields like medicine, agriculture, and remote sensing [26,27,28,30]. Deep learning techniques work best when there are noisy measurements, few labels, and an imperfect understanding of physics, all of which are also important in the context of astrophysical data analysis. As

such, deep learning techniques have now become important in modeling stellar populations and astrophysical transients.

Red supergiants (RSGs) have long been known to be principal type IIP supernovae progenitors, as established by direct detection in pre-explosion cataloged images and statistical analysis of neighboring supernovae [6,16,17]. The progenitor masses are known to be constrained within 8–25 $M_{\odot}$, but less luminous RSGs are still under-abundant in model predictions of stellar evolution theory, known as the "red supergiant problem" [6,17]. An indication of increased mass loss or different evolution before collapse is indicated [7,18].

Large-scale stellar surveys, such as Gaia Data Release 3, combined with near-infrared observations from 2MASS and optical imaging from Pan-STARRS1, have for the first time enabled population-level characterisation of evolved massive stars, including dusty and highly evolved RSGs that may be missed in optical-only samples [4,5,8,9,20]. In parallel, modern time-domain surveys have driven the adoption of machine learning for automated supernova classification, with strong performance achieved using post-explosion light curves and image cutouts [10,11,13,14]. However, these approaches remain limited to post-explosion observables and do not address progenitor behaviour prior to explosion.

Other recent works have investigated pre-explosion variability and circumstellar signatures as possible precursors, but their usefulness is limited by the sparsity of time domain coverage as well as observation selection effects. Multimodal machine learning offers a promising solution by enabling the joint analysis of heterogeneous data sources. Image–tabular fusion models have shown superior performance in several astrophysical tasks [23,24], yet their application to pre-explosion supernova progenitor prediction remains limited [10].

This work addresses this gap by proposing a multimodal learning framework that integrates pre-explosion imaging with astrophysical catalog parameters to estimate the explosion likelihood of red supergiants, complementing existing post-explosion classification pipelines.

3 Proposed Methodology

The aim of this research work will be to develop a predictive model capable of estimating the probability that a Red Supergiant (RSG) star may undergo core collapse and then explode as a supernova. In the implemented project, actual pre-explosion Pan-STARRS1 (PS1) imaging of established Type II progenitors and meticulously constructed negative RSG images are converted into 200×200 pixel patches, normalized, and passed through a ResNet-18 encoder for capturing 512-dimensional CNN feature representations along with a small set of handcrafted image statistics.

In parallel, highly incomplete columns are eliminated from the RSG catalog, and the remaining features are subjected to median imputation. By using alias matching to well-researched extreme RSGs, binary labels separate "normal" RSGs from a small high-risk subset. A 986-row, star-level training set that combines deep image embeddings with astrophysical parameters like position, photometry,

and effective temperature is created by combining these catalog entries with the derived image features. Stratified cross-validation is used to train a tuned Random Forest classifier with a PCA bottleneck over the CNN features, resulting in a probability of explosion (SN score) for each star that represents how similar it is to known pre-supernova progenitors in both image morphology and catalog properties.

Figure 1 provides a clear summary of the multimodal pipeline, illustrating how pre-explosion images and astrophysical catalog feature processing are fused and passed to the modeling framework to produce a calibrated supernova explosion probability score.

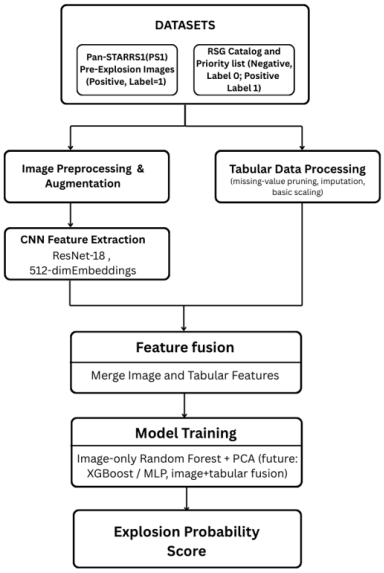

Fig. 1. Overview of the multimodal pipeline integrating PS1 progenitor imaging, RSG catalog data, CNN feature extraction, feature fusion, and Random Forest modeling to produce a calibrated explosion probability score.

3.1 Datasets

Pan-STARRS1 (PS1) Pre-Explosion Imaging. We selected pre-explosion images of known supernova progenitors (Types II, IIb, IIP, etc.) from the PS1 Image Server. Each target corresponds to a verified progenitor (e.g., SN2004A, SN2004et, SN2011dh, SN2005cs). Multi-band FITS images were obtained in the *g, r, i, z, y* bands. After handling server inconsistencies and invalid FITS data structures, a total of 135 validated single-HDU images were retained as positive examples (label 1).

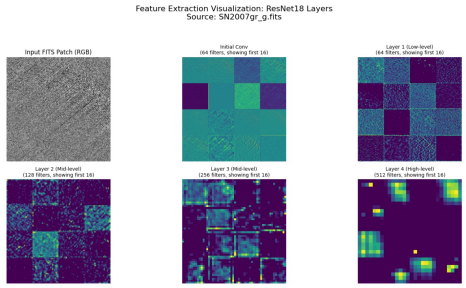

Fig. 2. Representative PS1 pre-explosion progenitor image (left) and intermediate activation maps from successive ResNet-18 layers (right).

Figure 2 illustrates a representative PS1 pre-explosion image alongside intermediate activation maps extracted from successive layers of the pretrained ResNet-18 encoder. For a typical young star-forming progenitor, intermediate activations were recorded across the network. Shallow layers are primarily sensitive to noise textures, while deeper layers respond to local gradients and suppress background structure, focusing instead on compact, high-level morphological characteristics near the stellar source. This demonstrates that the resulting 512-dimensional embedding captures a structured hierarchy of features suitable for fusion with tabular astrophysical measurements.

Augmented Supernova Progenitor Dataset. Each FITS image underwent centering to produce a 200×200 pixel patch, followed by data augmentation operations including rotations, flips, scaling, noise injection, and contrast adjustments. These transformations resulted in 95 usable augmented progenitor examples, which were encoded as RSG-style tabular rows and combined with the original PS1-derived features.

Red Supergiant (RSG) Catalog. The RSG catalog consists of 651 negative examples characterized by astrophysical parameters such as effective temperature ($T_{\rm eff}$), luminosity, radius, extinction, Gaia astrometry, and 2MASS photometry. The number of features decreased from 179 to 106 after columns with more than 80% missing values were eliminated. After applying median imputation to the remaining columns, 641 clean RSG entries with the label 0 were obtained.

Priority Progenitor List (Special Label = 1 Group). Ten high-risk RSGs with extreme mass loss or late-stage instability were included in the priority progenitor group (e.g., VY CMa, AH Sco, VX Sgr, μ Cep, S Per, NML Cyg). These objects were assigned a special designation of label 1. Together with the PS1 progenitor images, this group forms a total of 305 positive examples, enhancing model sensitivity to rare but physically significant progenitor behavior.

Both PCA-reduced CNN embeddings and manually created image statistics have a major impact on classification performance, according to feature importance analysis. Among the most important handcrafted features are brightness-related metrics like maximum pixel intensity, mean flux, and entropy. Brightness-related metrics such as maximum pixel intensity, mean flux, and entropy rank among the most influential handcrafted features. In parallel, PCA components encode complex morphological information learned by the CNN. These comple-

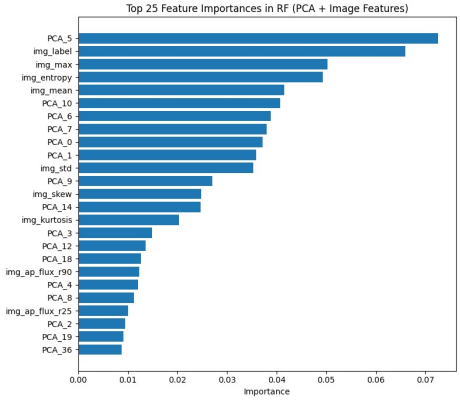

Fig. 3. Feature-importance visualization for PCA-reduced CNN embeddings and handcrafted image statistics.

mentary representations demonstrate that both low-level photometric detail and high-level learned features are critical for distinguishing high-risk RSGs from apparently normal counterparts.

3.2 Image Preprocessing Pipeline

The validation and preprocessing of each FITS image were performed using the `astropy` library. From each file, the first valid two-dimensional array was extracted. The resulting image was centered to a 200×200 pixel patch, normalized, and converted into an RGB representation. The images were then resized to 224×224 pixels and normalized using ImageNet statistics (mean $= [0.485, 0.456, 0.406]$), allowing compatibility with modern CNN architectures. All 135 progenitor and control RSG images were processed through this unified preprocessing pipeline.

3.3 Deep Feature Extraction (CNN Embeddings)

For deep feature extraction, a pre-trained ResNet-18 encoder was employed with the final fully connected layer removed and replaced by an identity mapping. This configuration produced 512-dimensional embedding vectors that capture both morphological and photometric structure information. In addition to the CNN embeddings, six handcrafted statistical features—including mean intensity, standard deviation, percentiles, entropy, and aperture flux measurements—were appended to the feature representation.

Batch inference was performed to generate a structured embedding dataset, which was stored in the file `cnnfeaturesposnegfull.csv`. Feature importance analysis using CART models indicates that both PCA-reduced CNN embeddings and manually crafted pre-explosion PS1 image statistics contribute significantly to classification performance. The results demonstrate that learned deep features capture morphological structure, while low-level photometric statistics encode flux-based characteristics. Together, these representations collectively encode

information relevant to late-stage stellar instability, confirming the hypothesis that ResNet-18 embeddings effectively capture transferable astrophysical features suitable for fusion with tabular catalog parameters in a multimodal modeling framework.

3.4 Tabular + Image Feature Fusion

The CNN-generated embeddings were transformed into a schema, which is consistent with the RSG, concatenated with the pre-processed RSG catalog, creating a combined training data set (`combined_training_dataset.csv`). The total number of samples in this dataset was 986, consisting of 634 variables, of which 681 were classified as negative, while 305 were classified as positive based on a binary categorization scheme. The proposed technique retrieves information from imaging, as well as attributes of the astrophysical data set.

3.5 Model Construction

The target of the proposed model will be to predict the likelihood of core collapse for a red supergiant (RSG) based on the pre-explosion observations that have been made. Due to the nature of the given observations being from different modalities with a lower number of confirmed progenitors being available and the imaging characteristics being of a high-dimensional form compared to the astrophysical variables, the task will not be an end-to-end learning task.

Each star is represented by two complementary feature sets that are fused into a unified representation. On the imaging side, pre-explosion FITS cutouts are processed using a pre-trained ResNet-18 encoder with the final classification layer removed. Residual learning enables stable feature propagation through the network,

$$h_i^{(b+1)} = h_i^{(b)} + F^{(b)}\big(h_i^{(b)}; \theta^{(b)}\big), \tag{1}$$

where $h_i^{(b)}$ is the feature map extracted from the residual block b. Although it seems like an ordinary skip connection, residual blocks retain information about low-level spatial details when refining high-level feature representations, which is crucial when training is conducted under limited data.

The final feature maps of the convolutional layers are transformed to a compact representation through global average pooling,

$$g_i(c) = \frac{1}{HW} \sum_{u=1}^{H} \sum_{v=1}^{W} h_i^{(L)}(c, u, v), \tag{2}$$

followed by a linear projection,

$$f_i^{\text{img}} = W_{\text{img}} g_i + b_{\text{img}}, \tag{3}$$

yielding a fixed-length (512-dimensional) image embedding. This embedding captures morphological and background structure in the stellar field while mitigating overfitting through dimensionality reduction and parameter-free pooling.

On the tabular side, each RSG is described by a curated set of astrophysical parameters, including effective temperature, luminosity, extinction, astrometric quantities, near-infrared photometry, colour indices, and variability indicators. Prior to fusion, tabular features are standardised as

$$\hat{z}_{i,j} = \frac{z_{i,j} - \mu_j}{\sigma_j}, \tag{4}$$

The embeddings for images and the standardized tabular features are then merged to create a common multimodal representation.

The embeddings of images and standardized tabular features are combined through concatenation into a single unified feature representation,

$$f_i = \left[f_i^{\text{img}} \parallel \hat{z}_i \right], \tag{5}$$

allowing the model to jointly exploit spatial morphology and physical stellar properties when estimating explosion likelihood.

Although a conceptual three-level labelling scheme is defined—confirmed progenitors, normal RSGs, and extreme high-risk RSGs—the current implementation adopts a binary formulation due to data limitations, with extreme cases folded into the positive class. The fused features are used to train interpretable tabular classifiers such as Random Forests or gradient-boosted trees, which can model non-linear interactions while providing feature importance estimates.

The explosion likelihood is computed as

$$s_i = w^\top f_i + b + \alpha r_i, \tag{6}$$

and transformed into a probability using the sigmoid function,

$$p_i = \sigma(s_i) = \frac{1}{1 + e^{-s_i}}. \tag{7}$$

This allows for probabilistic ranking instead of binary classification, as is more suitable for prioritiation done on observations.

The problem of class imbalance can be handled by the weighted binary cross entropy loss function.

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} w_i \Big[y_i \log p_i + (1 - y_i) \log(1 - p_i) \Big], \tag{8}$$

where higher weights are assigned to rare progenitor-like objects. Stratified cross-validation is employed to obtain reliable performance estimates while preventing information leakage.

One of the main design objectives is interpretability. While ablation studies evaluate the relative contribution of astrophysical and image-derived features, tree-based models allow feature importances to be examined. Each RSG receives a continuous explosion probability score from the final model, which is meant to serve as a relative risk ranking to direct subsequent observations rather than a final forecast.

4 Datasets and Feature Design

The dataset integrates three primary sources: pre-explosion Pan-STARRS1 (PS1) imaging, a Gaia-based red supergiant (RSG) catalogue, and a curated set of high-risk RSGs exhibiting extreme mass loss and variability. PS1 provides multi-epoch imaging in five broadband filters over a 3π steradian sky area with uniform calibration, enabling deep stacked cutouts around each target [8,9]. Gaia Data Release 3 contributes high-precision astrometry and low-resolution spectrophotometry, from which effective temperatures, luminosities, and extinctions are derived, while near-infrared surveys supply complementary fluxes and colours sensitive to cool photospheres and dusty envelopes [20,2,19].

High-risk RSGs are identified from the literature based on strong infrared excess, irregular photometric variability, and evidence of enhanced mass loss, and are treated as an intermediate class occupying the late pre-supernova regime even without direct explosion confirmation [1,21,22]. Confirmed progenitors with available pre-explosion imaging anchor the positive label and ensure that the model representation includes stars known to have exploded.

4.1 Model Training Strategy and Optimization

To ensure robust evaluation, the multimodal model is trained using stratified cross-validation that preserves class imbalance, with weighted sampling to counter the dominance of normal RSGs. Because there are few confirmed progenitors, regularization strategies like dropout, batch normalization, and early stopping are used to reduce overfitting.

Grid search is used to adjust the Random Forest baseline's hyperparameters for tree depth, impurity criteria, class weights, and minimum sample thresholds. After comparison with autoencoders and sparse feature selection, dimensionality reduction using PCA is chosen for the multimodal pipeline because it offers more stability under cross-validation.

Because the model outputs a continuous explosion likelihood, calibration is treated as a critical component. Isotonic regression is adopted over Platt scaling, yielding improved alignment between predicted probabilities and astrophysical ordering of the label groups. Probability histograms exhibit a smooth transition across labels, indicating that the model captures a continuum of pre-supernova instability rather than discrete classes, which is advantageous for ranking and prioritisation in follow-up observations.

5 Results and Discussion

The multimodal MPN-RSG neural network is trained on a carefully selected dataset of 676 objects, which includes 580 normal red supergiants (RSGs), 80 high-risk RSGs, and 16 confirmed progenitors with accessible pre-explosion images. Each objects is described by a set of astrophysical properties, along with a 512-dimensional representation based on the ResNet-18 neural network

extracted from Pan-STARRS1 images whenever available. For objects without images, a "missing-modality" approach enables the neural network to train using only the astrophysical data, leveraging the power of images during simultaneous availability.

In comparison to a Random Forest trained only in a tabular manner on the same set of astrophysical attributes, a multimodal AI system has a greater level of success. In regard to accuracy, macro F1-score, and precision, a value of 0.90, 0.88, and 0.89 is obtained, with an area under an ROC curve measuring 0.91. These gains indicate that incorporating convolutional image features improves both class separation and the stability of explosion likelihood estimates.

To create an image-only baseline, manually created image descriptors are added to ResNet-18 embeddings. Principal component analysis reduces the 512-dimensional embeddings within each cross-validation fold to 64 components. To reduce dimensionality and overfitting, these components are concatenated with 16 features that capture flux distribution, contrast, and shape. The number of usable positive instances is increased by treating individual FITS cutouts as distinct star-level samples rather than aggregating across filters.

The image-only approach relies on a 500-tree Random Forest with class weights, median imputation, and eight-fold stratified cross-validation. When there is a class imbalance, F1-score optimization is used as a decision threshold to strike a balance between recall and precision. Although the baseline shows a fair level of discrimination, it is always worse than the multimodal approach, which highlights the relevance of integrating imaging and astrophysical knowledge.

Image-only model performance metrics averaged over the folds are provided in Table 1.

Table 1. Performance Evaluation of the Image-Only Classification Model

Evaluation Metric	**Mean ± Std. Dev.**
Accuracy	0.782 ± 0.022
Precision	0.946 ± 0.063
Recall	0.311 ± 0.061
F1-score (threshold = 0.5)	0.466 ± 0.075
Area Under ROC Curve (AUC)	0.666 ± 0.032
Optimal F1 Threshold	0.551
F1-score at Optimal Threshold	0.508 ± 0.031

Probability histograms show a smooth progression of predicted scores, indicating a continuous measure of pre-supernova instability rather than sharply separated classes. High-probability candidates align with independently proposed progenitors, while low-probability objects cluster among normal RSGs. This behaviour enables effective ranking for follow-up observations, with low prediction variance indicating good generalisation despite limited image-based data.

6 Conclusion and Future Work

This study demonstrates that a multimodal learning framework can effectively identify likely supernova progenitors by jointly analysing pre-explosion imaging and red supergiant (RSG) astrophysical parameters. The suggested method goes beyond conventional post-explosion categorization and moves predictive attention toward progenitor-level evaluation by combining convolutional neural network embeddings taken from archival images with physically significant stellar properties. Despite the challenges posed by a small number of confirmed progenitor cases and limited computational resources, the methodology offers a thorough and repeatable pathway from FITS validation and picture augmentation to feature extraction, fusion, and calibrated probability estimation.

The results demonstrate that combining structural information from imaging with catalog-based stellar attributes improves explosion-likelihood estimation and enables conservative, high-confidence prioritization of possible progenitors for follow-up observations. This multimodal approach offers a feasible balance between model complexity, interpretability, and performance, making it suitable for short-term implementation in monitoring and survey-driven processes.

Future research will concentrate on improving catalog completeness, adding more surveys, and standardizing the underlying datasets through deeper multi-band imaging. More expressive fusion architectures, refined deep networks, and physics-informed or time-aware models that may incorporate uncertainty estimates and temporal evolution will all be investigated as methodological extensions. These advancements will increase data-driven frameworks' capacity to support dependable early-warning systems for star explosions.

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